

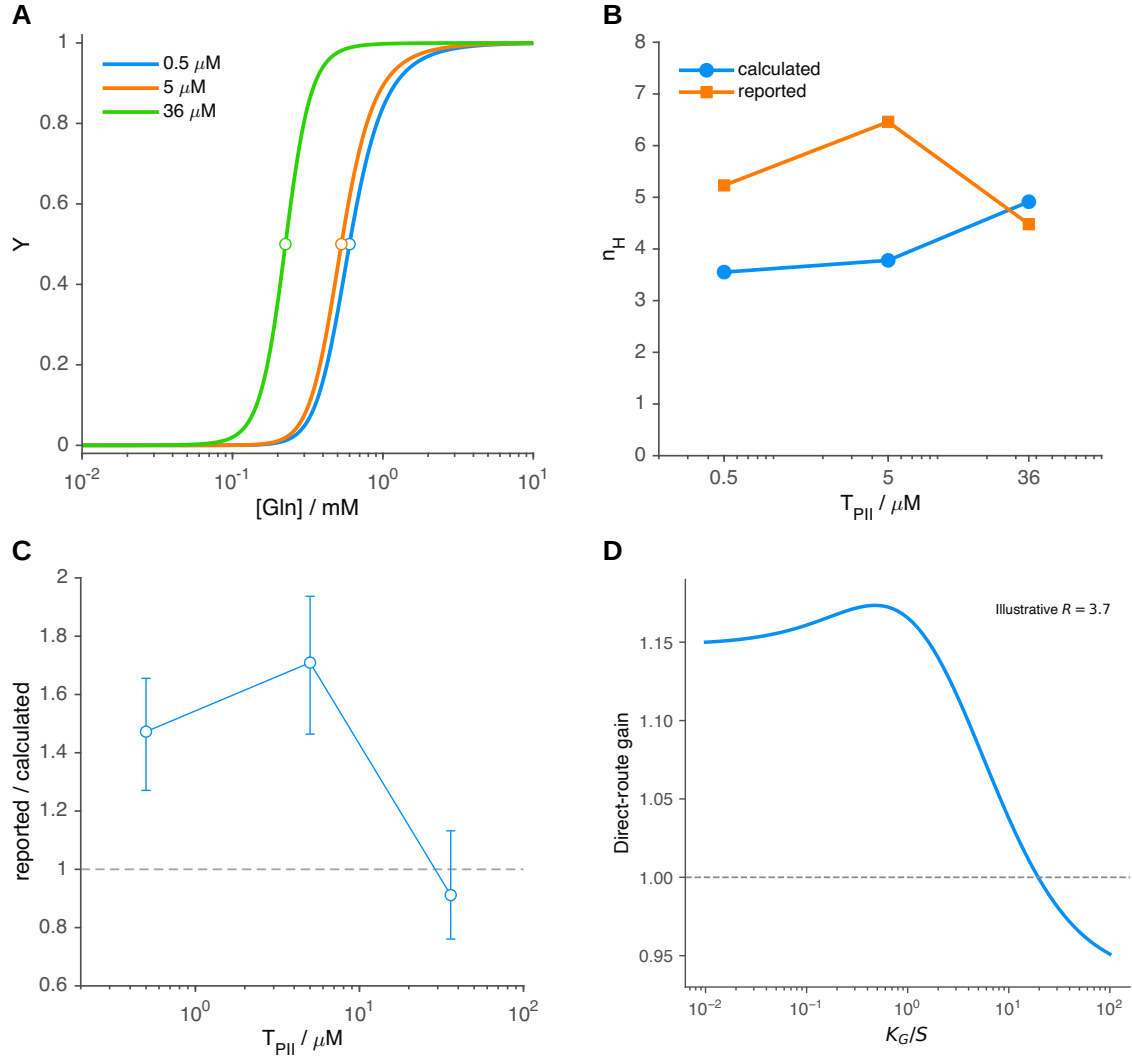


Figure S9: Conditional accounting of cascade sensitivity and a direct glutamine route. (A) Calculated downstream responses normalized between their limiting plateaus. Points mark calibrated reported midpoints; curves are model outputs, not experimental traces. (B) Calculated response-range coefficients and reported cascade coefficients at three PII concentrations. PII concentrations of 0.5, 5, and 36 micromolar correspond to UTase/UR concentrations of 0.05, 0.5, and 0.8 micromolar. (C) Ratios of reported to calculated coefficients. Error bars span central 95% ranges from upstream coordinate perturbations only; connected points aid reading and do not establish a non-monotonic mechanism. (D) Gain produced by the illustrative direct activation law with $R=3.7$ while retaining the downstream scale calibrated before adding the route. The gain spans approximately 0.95–1.17 and does not supply the middle-condition discrepancy in this specified construction.